TEL200 Introduction to Robotics

Chapter 4: Mobile Robots

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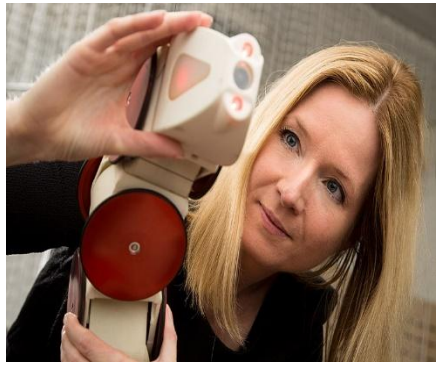
Mobile robots

- Walking Robot Lab
- Mobile robots: locomotion
- Car-like mobile robots
  - Unicycle and bicycle models
- Non-holonomic constraints
- Omnidirectional vehicles
- Flying robots
  - Quadrotor model
  - Quadrotor control

Mobile robots: various means of locomotion

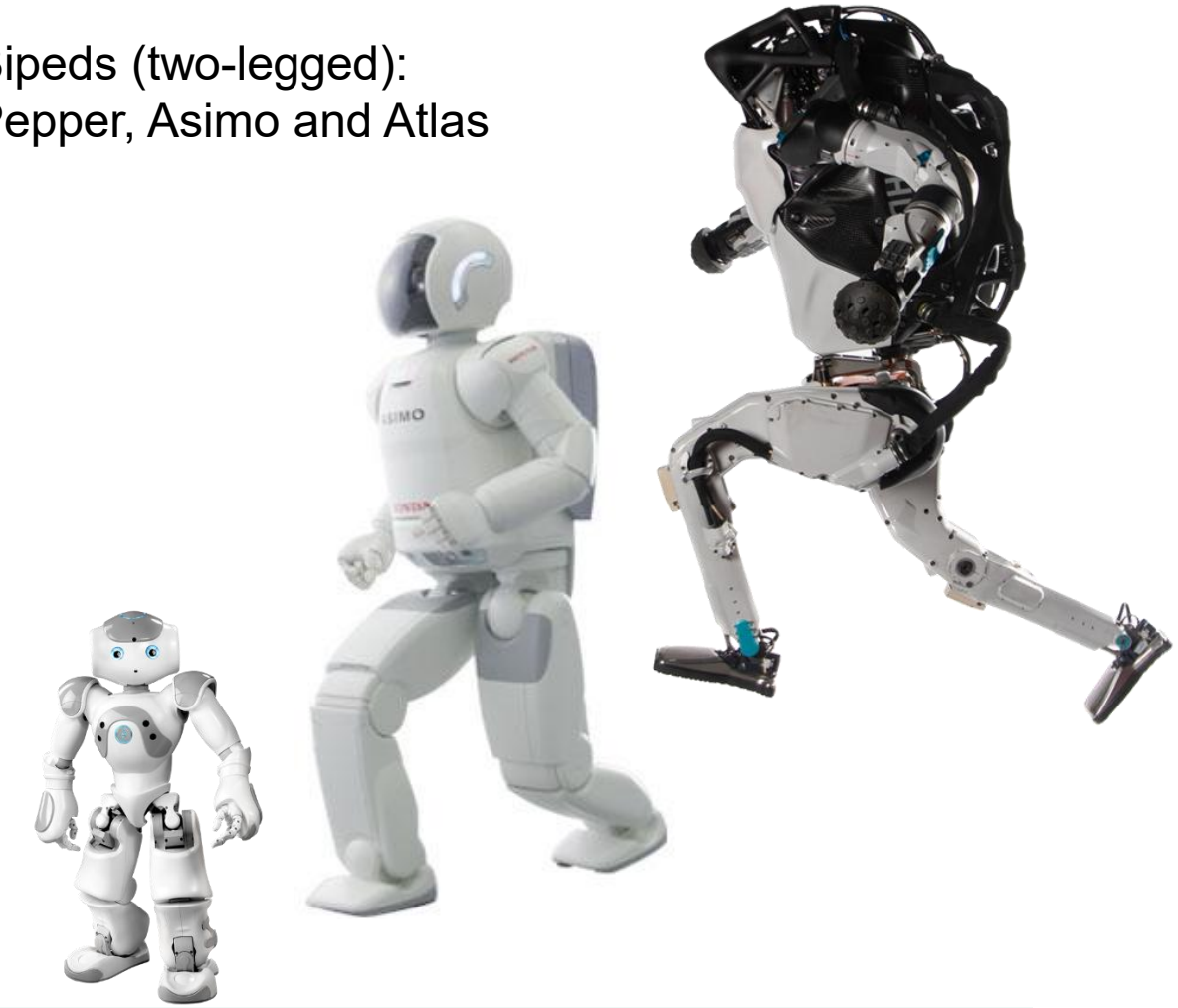
- Wheels
- Tracks
- Legs
- Flying
- Water surface
- Under-water
- Snake-like
- ….



Legged locomotion

- Point contacts between robot and ground
- Pros
  - Potential for handling rough terrain
  - Only contact points need to be OK, ground in between does not matter
- Cons
  - Mechanically complex
  - Power hungry

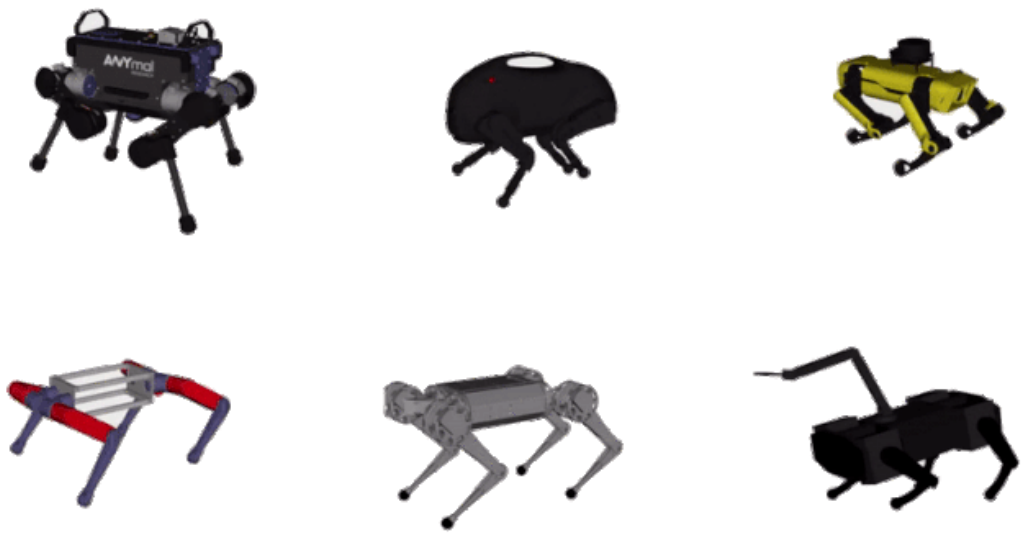
Bipeds (two-legged):
Pepper, Asimo and Atlas



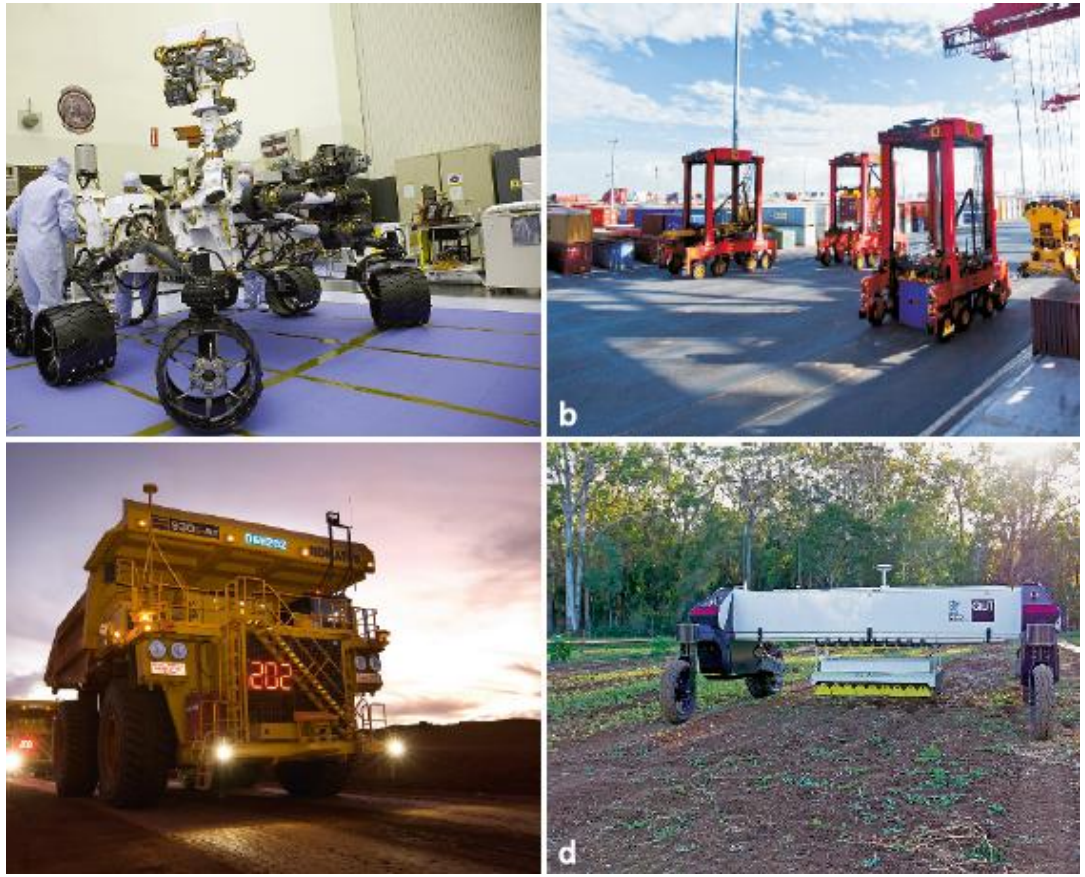
Legged locomotion

- Point contacts between robot and ground
- Pros
  - Potential for handling rough terrain
  - Only contact points need to be OK, ground in between does not matter
- Cons
  - Mechanically complex
  - Power hungry

Quadrupeds – statically stable:
ANYmal, Spot, …



Wheeled locomotion



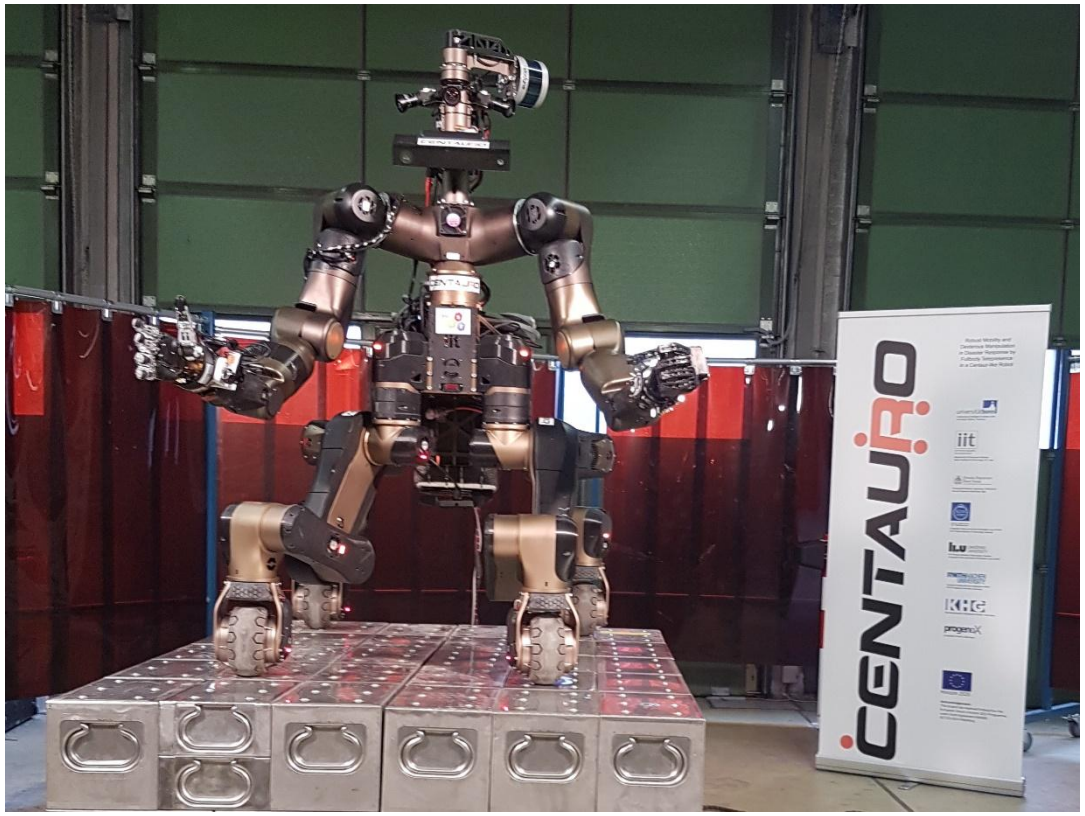
- Most popular means of locomotion
- Simple to implement, highly efficient
- Bigger wheels gives better handling of rough terrain
- Wheeled robots typically designed so that balance is not an issue

Locomotion: Skid/differential steering

- Mostly for outdoor platforms
- Wheels or tracks
- Turn by applying different speed to wheels
- Skidding/slipping makes it hard to predict motion
- Extremely energy inefficient when friction is high

Locomotion: Wheel-leg combos

- H2020 Project «Centauro»



- Hyundai walking-car «Elevate»

Video: Wheel-leg combo prototype

Locomotion: Flying - Unmanned Aerial Vehicles (UAVs)

- Two main types
  - Fixed-wings
  - Multi-rotor (rotary-wing)
- Student-groups discussions:
  - Pros and cons for these two types of UAVs

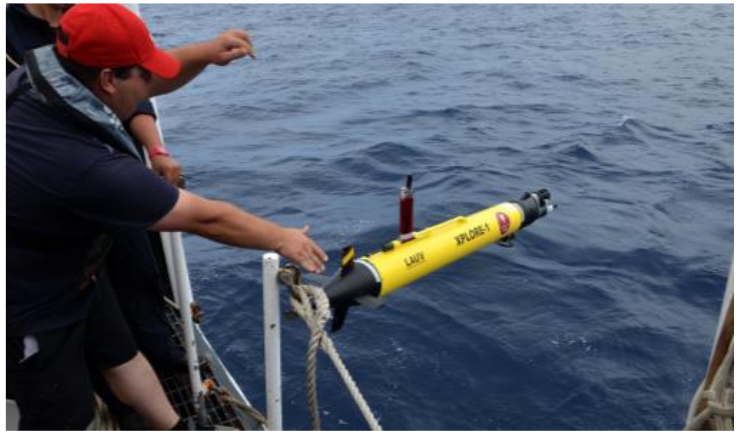
Locomotion: Pros and cons with fixed wing vs. multi-rotor

- Fixed-wings:
  - Can glide. Moves fast
  - Longer range
  - Cannot stand still
  - Need larger area for start/stop

- Multi-rotor:
  - Can stand still (“hover”). Easy to change height
  - Can fly close to ground
  - Start/stop on small area
  - Limited range

Video: Zipline delivery in Rwanda with fixed-wing UAVs

Locomotion: surface- and underwater robots



Mobile Robots – Locomotion and configuration space examples

Train

1 - DoF

$q = x$

Hover

3 - DoF

$\boldsymbol{q} = (x, y, \vartheta,)$

Helicopter

6 - DoF

$\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$

Fixed-wing aircraft

6 - DoF

$\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$

Quadrotor

6 - DoF

$\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$

Car

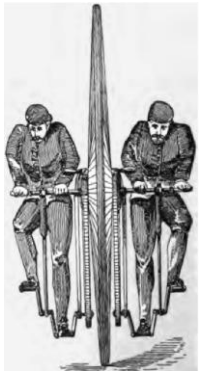
3 - DoF

$\boldsymbol{q} = (x, y, \vartheta,)$

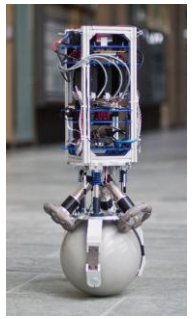
Mobile robots – RVC Chapter 4

- Wheeled
  1. Car-like (bicycle/unicycle)
  2. Differential drive
  3. Omnidirectional

$$\dot{x} = v\cos\theta$$
$$\dot{y} = v\sin\theta$$
$$\dot{\theta} = \frac{v}{L}\tan\gamma$$

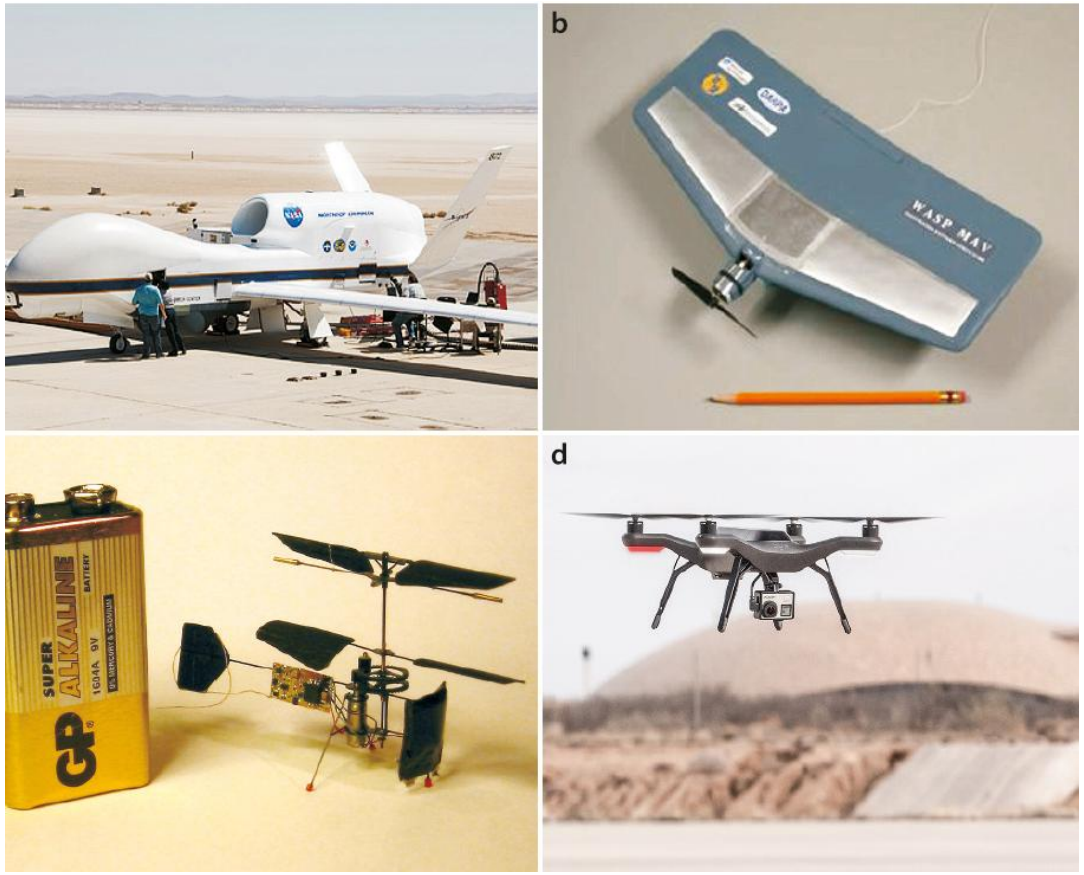


$$\dot{x} = v\cos\theta$$
$$\dot{y} = v\sin\theta$$
$$\dot{\theta} = \frac{v_\Delta}{W}$$



Mobile robots – RVC Chapter 4

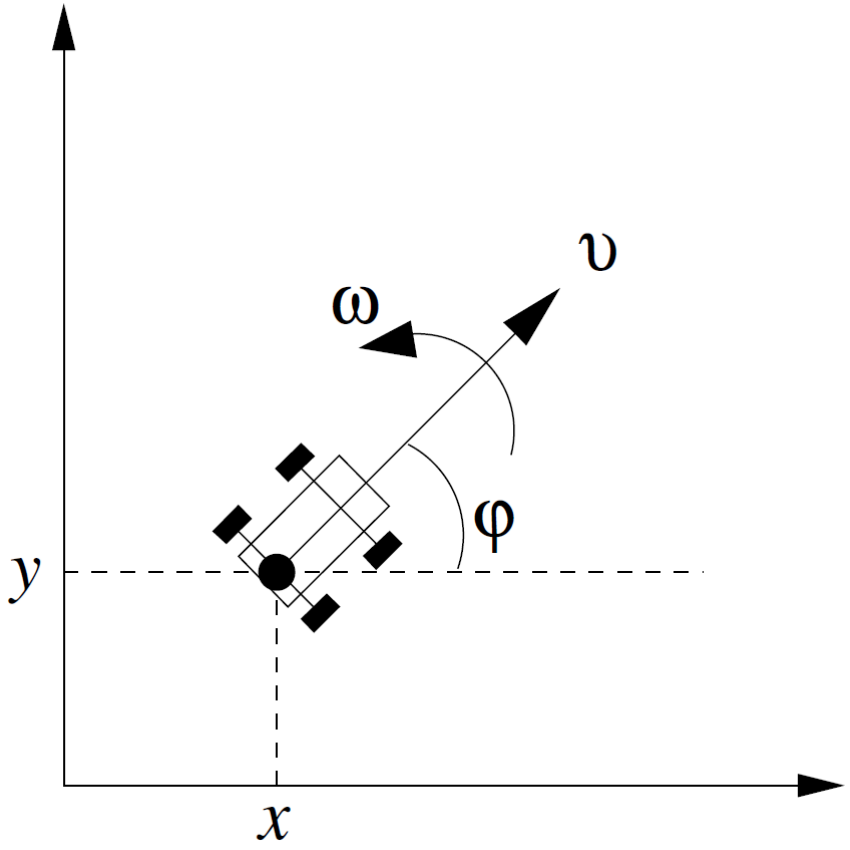
- Wheeled
  1. Car-like (bicycle/unicycle)
  2. Differential drive
  3. Omnidirectional
- Flying
  - Quadrotor
- Non-holonomic constraints



Car-like robots: Unicycle car model

- 3 DoF $q = (x, y, \varphi) \in \Re^2 \times S^1$
- 2 actuators: (v, ω)
- Path smoothness: C

$$
\begin{aligned}
\dot{x} &= v \cos \varphi \\
\dot{y} &= v \sin \varphi \\
\dot{\varphi} &= \omega .
\end{aligned}
$$



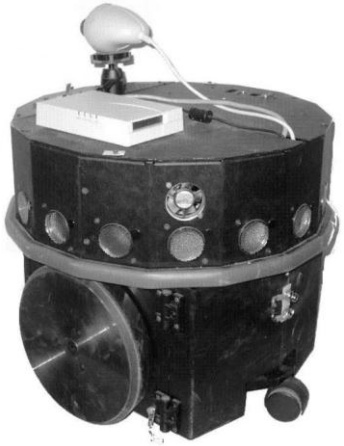
Car-like robots: Differentially-steered vehicle

- Unicyle model have no mechanically complex steerable wheels (Acherman steering mechanism)
- Instead uses two driven wheels + castor wheel(s)

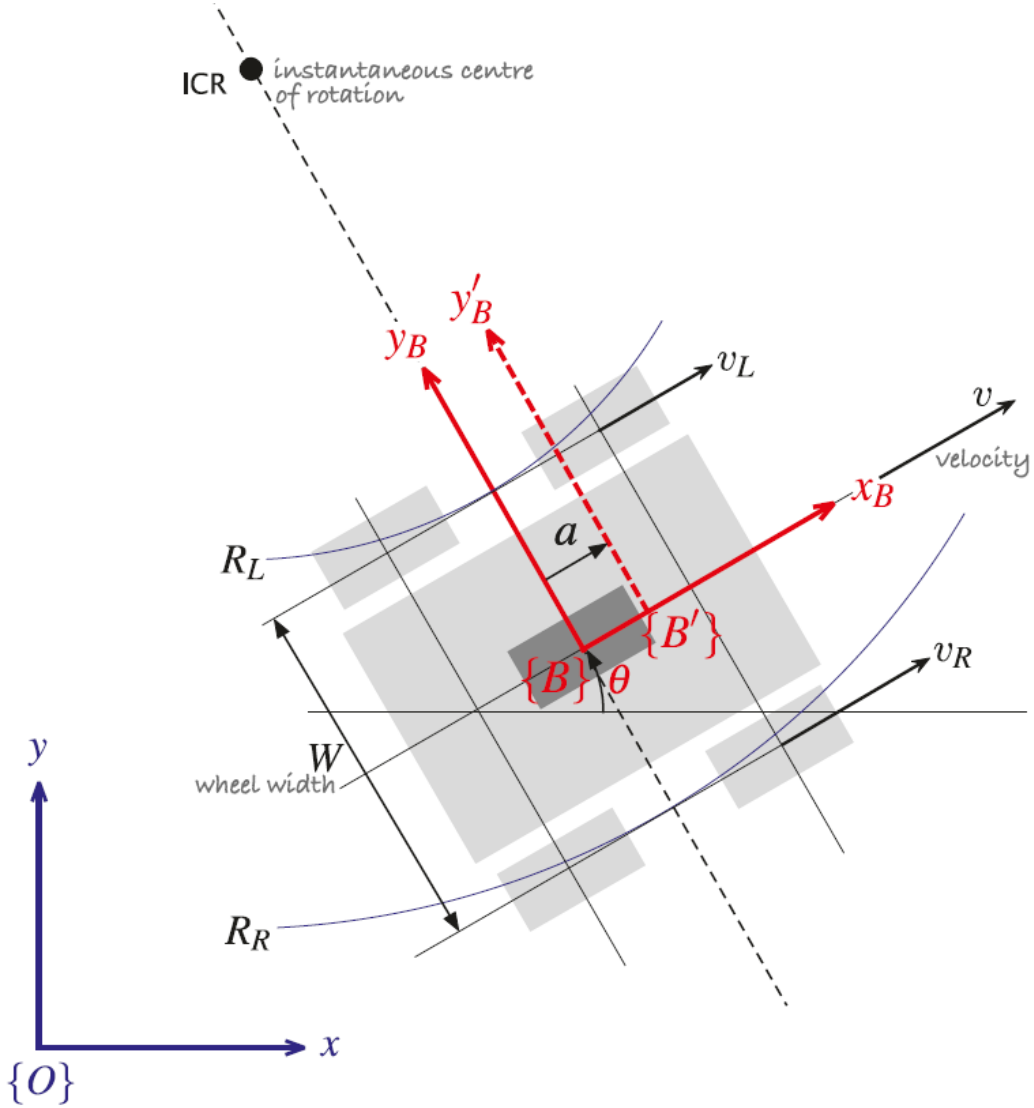
$$\dot{x} = v\cos\theta$$

$$\dot{y} = v\sin\theta$$

$$\dot{\theta} = \frac{v_\Delta}{W}$$

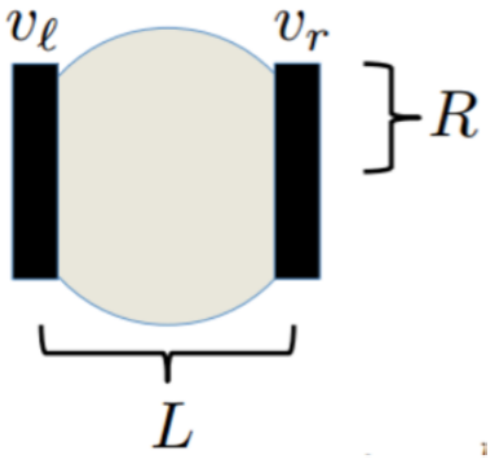


where $v = ½(v_R + v_L)$ and $v_\Delta = v_R - v_L$



* Photo courtesy: Nomadic Technologies Scout

Differentially-steered vehicle: unicycle model



$$\dot{x} = \frac{R}{2}(v_r + v_l)cos\theta$$
$$\dot{y} = \frac{R}{2}(v_r + v_l)sin\theta$$
$$\dot{\theta} = \frac{R}{L}(v_r - v_l)$$

Differential

$$\dot{x} = Vcos\theta$$
$$\dot{y} = Vsin\theta$$
$$\dot{\theta} = \omega$$

Unicycle

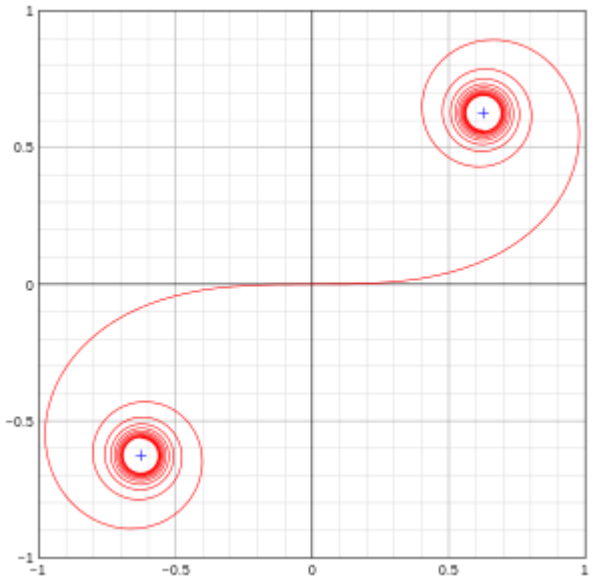
$$V = \frac{R}{2}(v_r + v_l) \Rightarrow \frac{2V}{R} = (v_r + v_l)$$

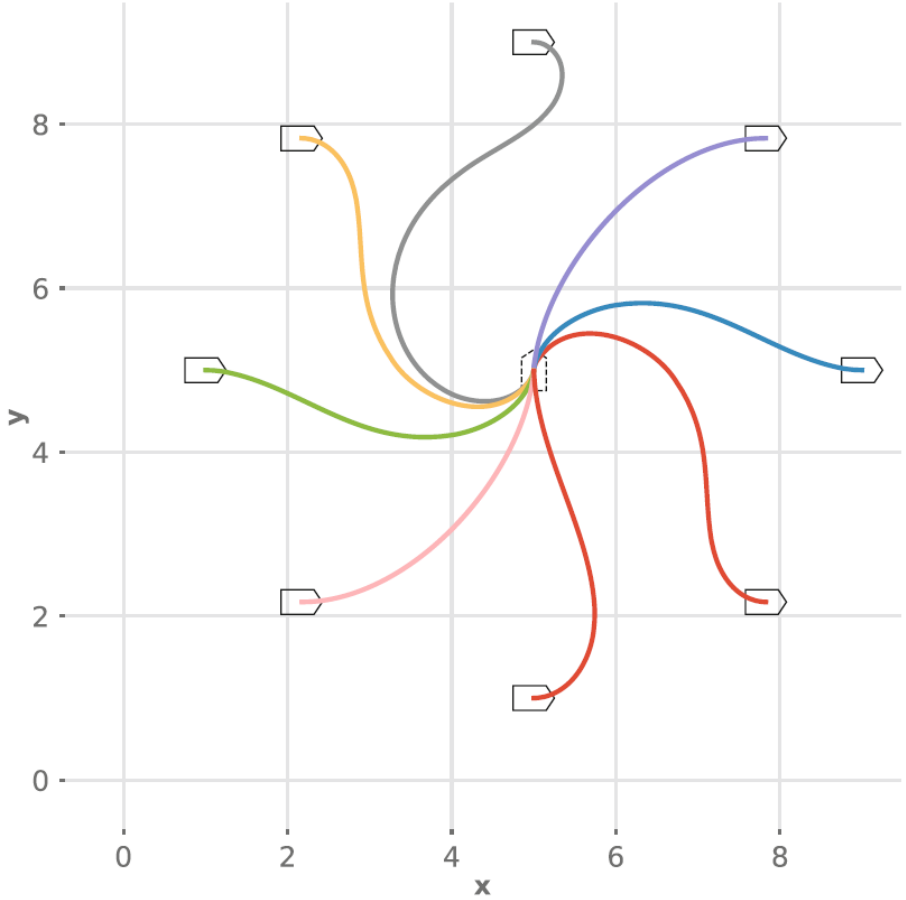
$$\omega = \frac{R}{L}(v_r - v_l) \Rightarrow \frac{\omega L}{R} = (v_r - v_l)$$

$$v_r = \frac{2V + \omega L}{2R}$$
$$v_l = \frac{2V - \omega L}{2R}$$

Car-like robots: paths with continuous curvature

- **Clothoid**: curvature changes linearly with its curve length
- Roads are normally curved as clothoids

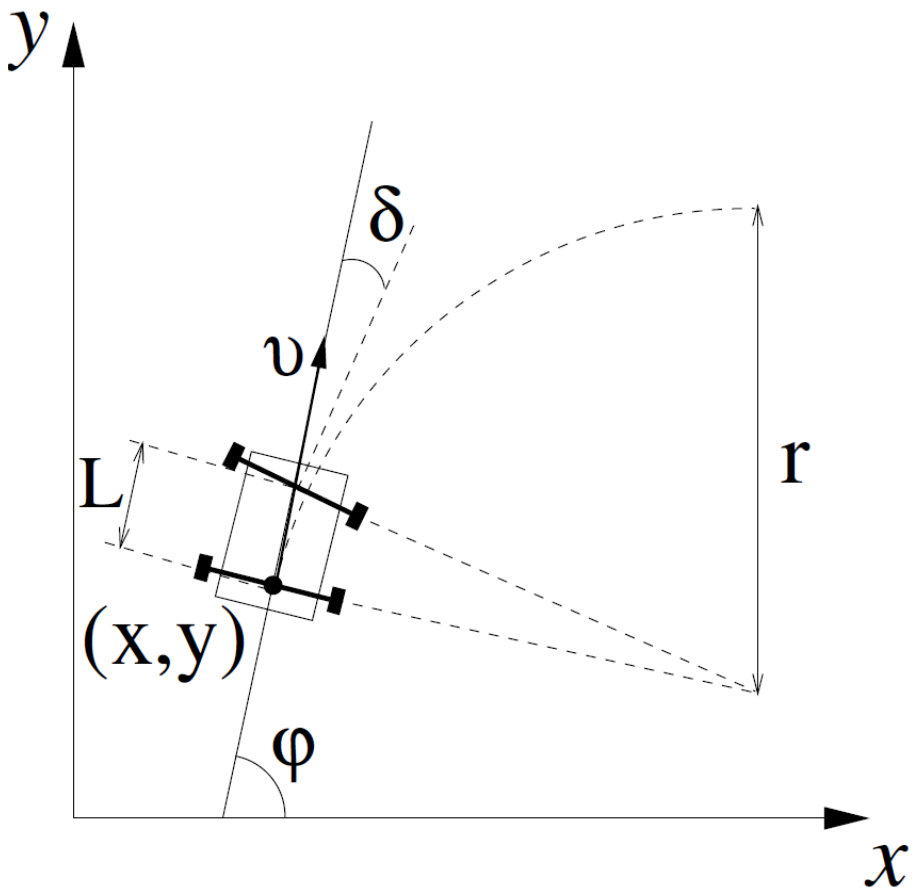




Car-like robots: Bicycle car model

- 3 DoF q = (x,y, φ) ∈ $\Re^2$ x S^1
- 2 actuators: (v, δ)
- Path smoothness: C^1

$$
\begin{aligned}
\dot{x} &= v \cos \varphi \\
\dot{y} &= v \sin \varphi \\
\dot{\varphi} &= \frac{v}{L} \tan \delta.
\end{aligned}
$$



Bicycle vs. Unicycle model (Exercise 4.10)

- Bicycle modelled, Ackerman steered robots are controlled by (v,γ)
- Unicycle modeled, differentially steered robots are controlled by V_L and V_R
- The relation between them is as follows:

$$v = \frac{V_R + V_L}{2}$$

$$\gamma = \arctan\left(\frac{L(V_R - v_L)}{vW}\right)$$

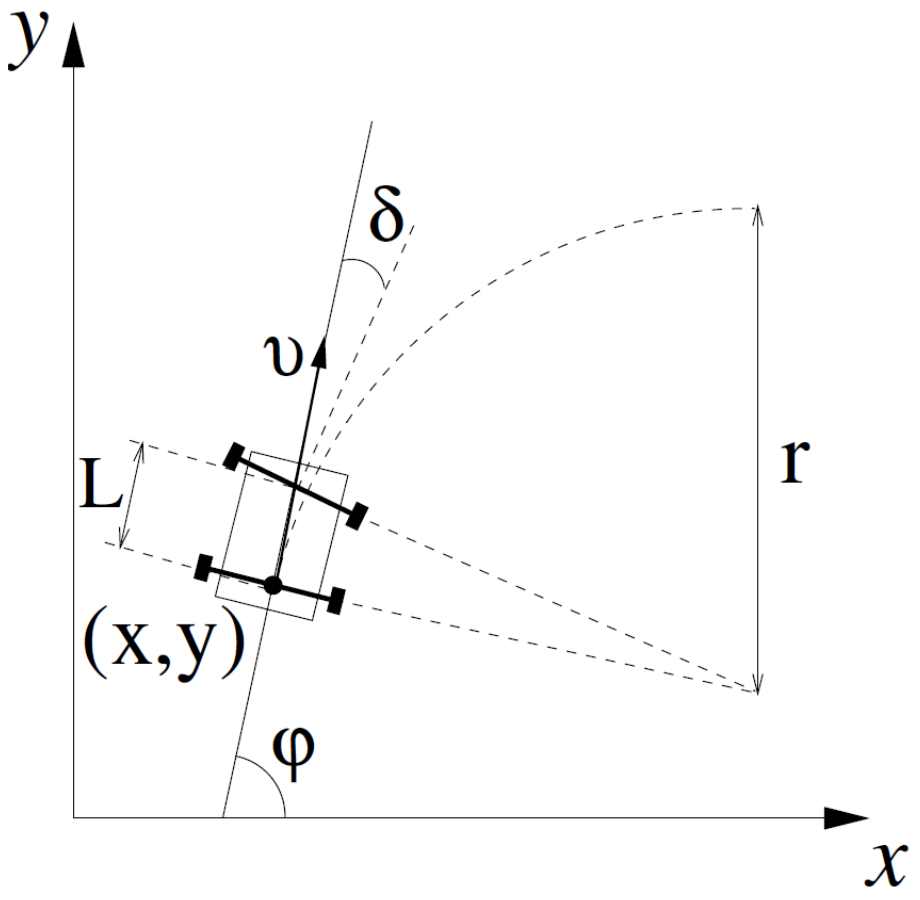
$$V_L = v - \frac{vW}{2L}\tan\gamma$$

$$V_R = v + \frac{vW}{2L}\tan\gamma$$

Car-like robots: Front-wheeled car model

- 4 DoF q= (x,y, φ, δ)
- 2 actuators: (v, ω)
- Path smoothness: C^2

$$
\begin{aligned}
\dot{x} &= v \cos \varphi \\
\dot{y} &= v \sin \varphi \\
\dot{\delta} &= \omega \\
\dot{\varphi} &= \frac{v}{L} \tan \delta
\end{aligned}
$$



Car-like robots: summary

Model name	Unicycle Robot Model	Car-like Model	Front-wheeled Car
System Dynamics	$\dot{x} = v\cos\varphi$ $\dot{y} = v\sin\varphi$ $\dot{\varphi} = \omega$	$\dot{x} = v\cos\varphi$ $\dot{y} = v\sin\varphi$ $\dot{\varphi} = \frac{v}{L}\tan\delta$	$\dot{x} = v\cos\varphi$ $\dot{y} = v\sin\varphi$ $\dot{\delta} = \omega$ $\dot{\varphi} = \frac{v}{L}\tan\delta$
Path Smoothness	C	C^1	C^2

Car-like robots: Assumptions and Constraints

Assumption 1 (Rolling Without Slipping) *It is customary to assume that the robot wheels do not slip. This assumption is legitimate in all moderate-speed scenarios.*

$$\dot{x} = v \cos \varphi \qquad \text{and} \qquad \dot{y} = v \sin \varphi.$$

Eliminating v, we get

$$\dot{x} \sin \varphi - \dot{y} \cos \varphi = 0.$$

Assumption 2 (Maximum Steering Angle) *We further assume that we have a limitation on the maximum steering angle* $|\delta| \leq \delta_{\max}$, *meaning that the turning radius for our car is lower bounded.*

How do these two assumptions constrain the kinematics of the robot?

Car-like robots: Assumptions and Constraints

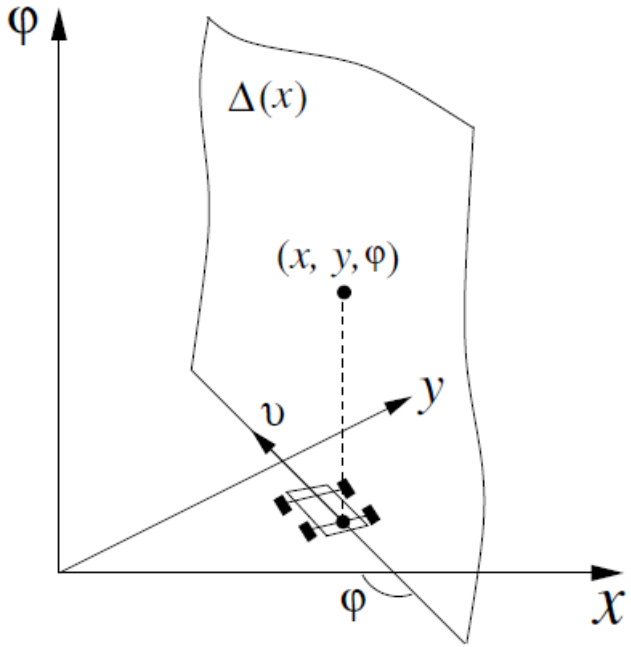
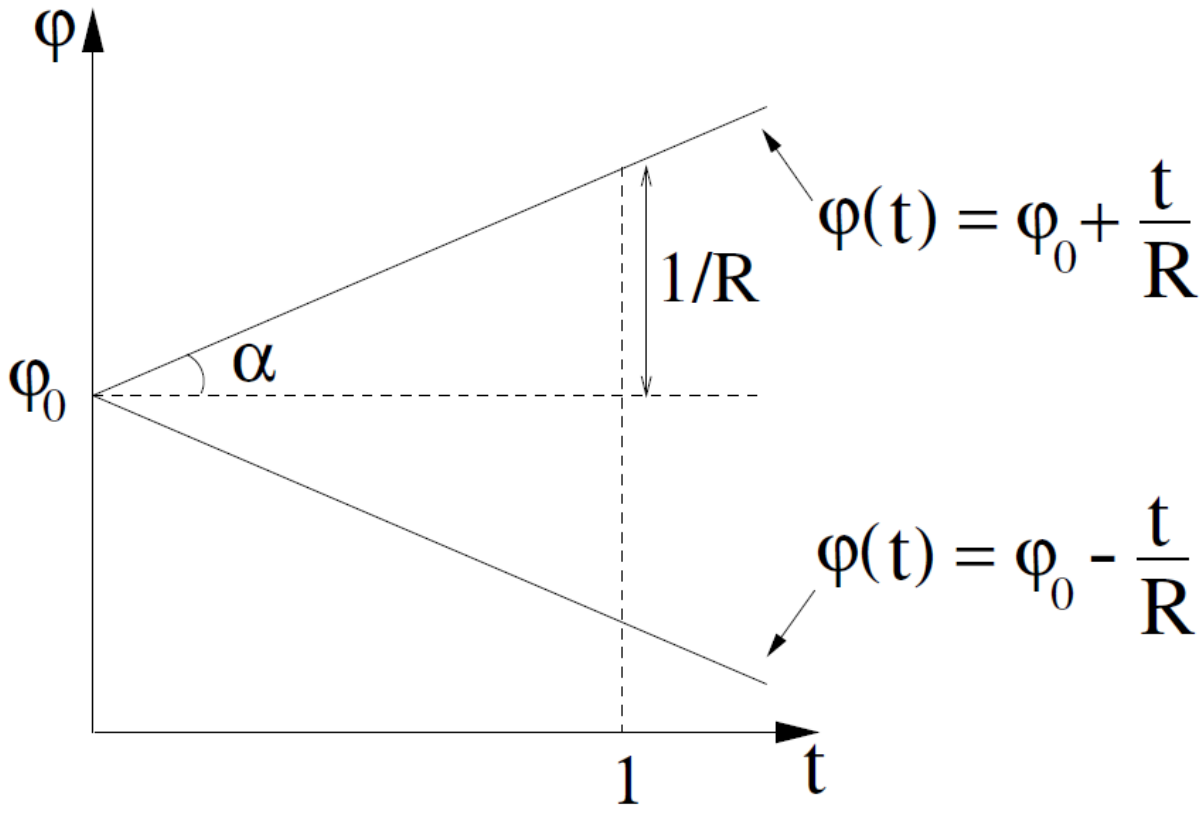
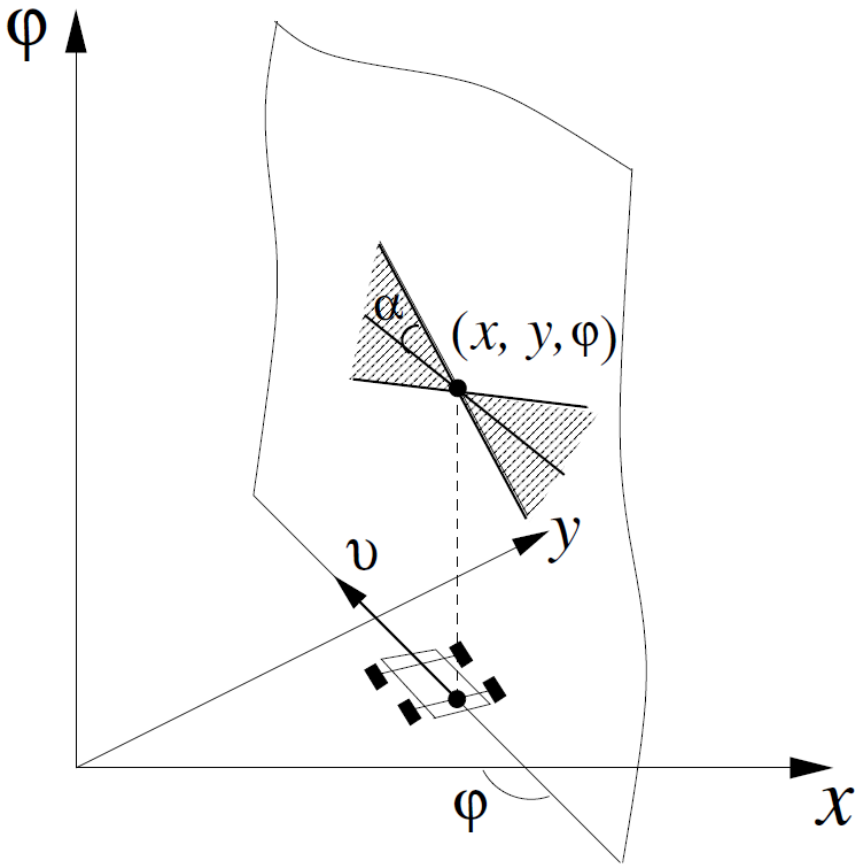


Figure 2.5: In the presence of the "rolling without slipping" assumption, the velocity vector $(\dot{x}, \dot{y})$, must lie in the direction of v and is therefore restricted to the 2-dim hyper surface $\Delta(x)$.

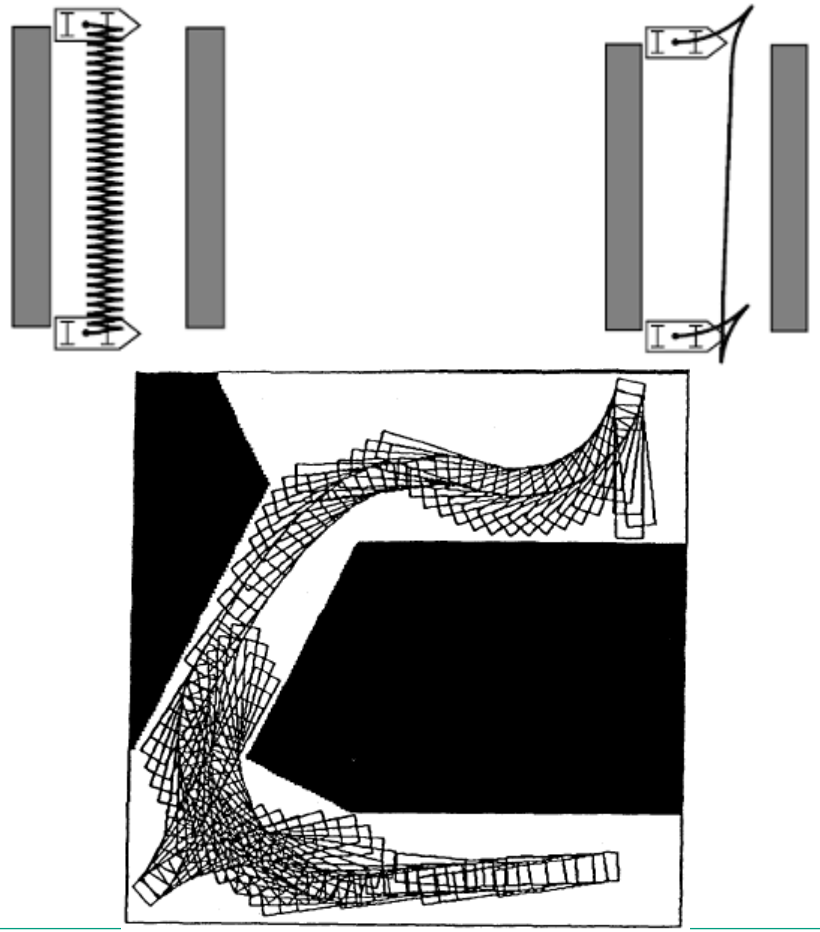
Car-like robots: Assumptions and Constraints





Car-like robots: Non-holonomic constraints

- Under-actuated: Fewer controls than dimensions in configuration space
  - Fewer actuators (less weight & cost)
  - Design simplicity
  - Convenience (think about driving a car with 3 controls!)
- “A vehicle with one or more velocity constraints, due to under-actuation or a rolling constraint, is referred to as a nonholonomic system”



Robot configuration space characteristics: summary

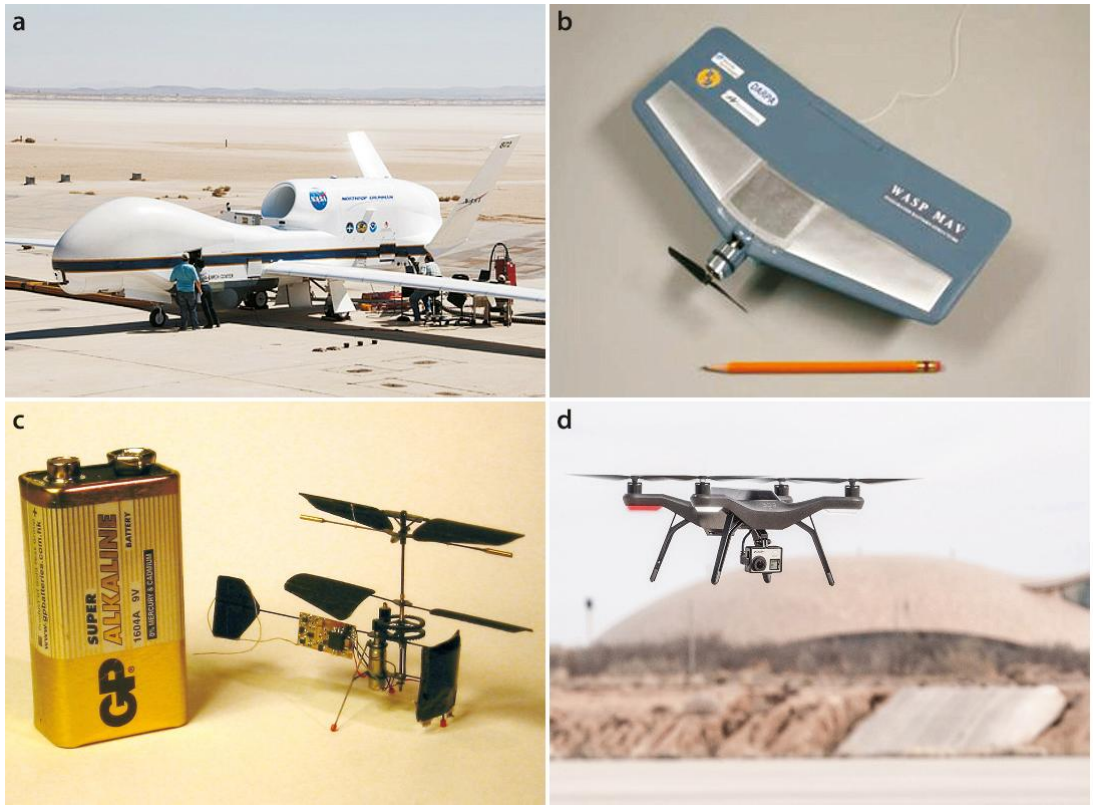
	dim $\mathcal{C}$	Degrees of freedom	Number of actuators	Actuation	Rolling constraints	Holonomic
Train	1	1	1	full		✓
2-joint robot arm	2	2	2	full		✓
6-joint robot arm	6	6	6	full		✓
10-joint robot arm	10	10	10	over		✓
Hovercraft	3	3	2	under		
Car	3	2	2	under	✓	
Helicopter	6	6	4	under		
Fixed wing aircraft	6	6	4	under		
DEPTHX AUV	6	6	6	full		✓

Omnidirectional vehicles

- Omnidirectional motion: Alternative wheel designs eliminate the non-holonomic, «rolling without slipping» constraint, e.g., «Swedish wheel»
- Requires 3+ suitably arranged Mecanum wheels, e.g., KUKA youBot or OmniMove

Flying robots - Unmanned Aerial Vehicles (UAVs)

- Main differences with ground vehicles
  - 6 DoF: translation and rotation
  - Actuated by forces, torques and accelerations rather than velocities: dynamic- rather than kinematic model
- Underwater robots: “fly through water”
  - Upward buoyancy force
  - Significant drag forces
  - Heavier



ProxDynamics/FLIR/ Black Hornet 3

- Provide Situational Awareness, Beyond visual LOS (BVLOS)
- Weight: <33g, 1.3 kg in total
- Total length: 168 mm
- 2 EO cameras and IR sensors
- Flight time: 25 min, range: 2km
- Max. speed: 6 m/s
- Wind tolerant op to gusts of 10 m/s
- Temperature range: -10°C to +43°C

Student group discussion

- Q: What is a drone?

Robotic Evolution – Drones

Robotic Evolution - Drones

- See video:
- https://www.britannica.com/video/182700/origins-diversification-drones-vehicles-devices-models

Quadrotor model 1/3

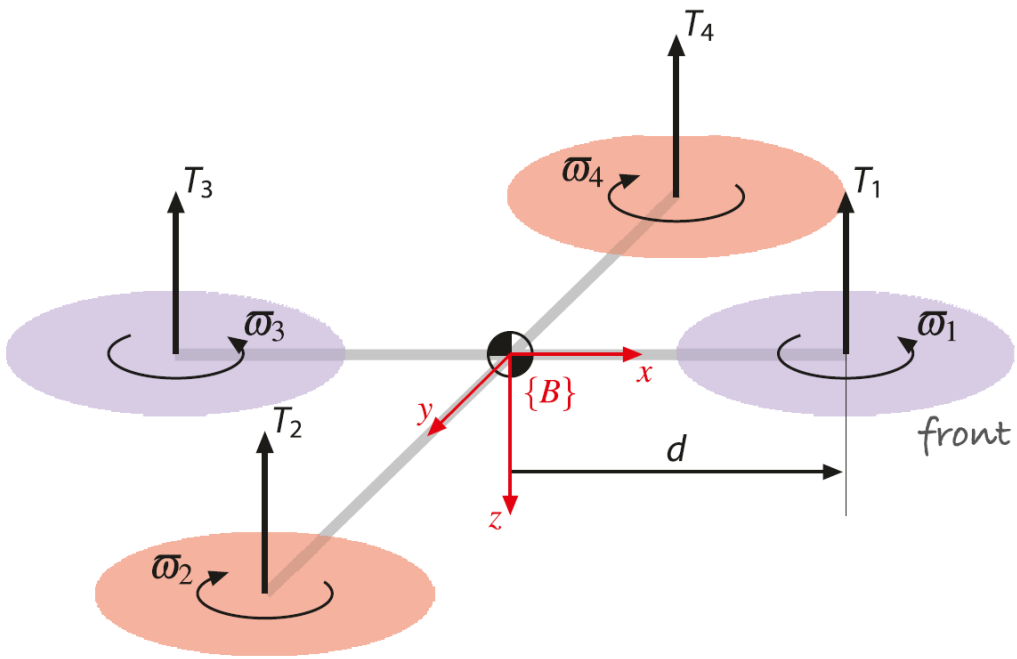
- The body frame {*B*} is attached to the vehicle with origin at the CoM
- *z*-axis downward (aerospace convention)
- Upward thrust: $T_i = b\varpi_i^2, \quad i = 1, 2, 3, 4$
- **Translational dynamics** (Newton's second law):

Aerodynamic drag

$$m\dot{\boldsymbol{v}} = \begin{pmatrix} 0 \\ 0 \\ mg \end{pmatrix} - {}^0\boldsymbol{R}_B \begin{pmatrix} 0 \\ 0 \\ T \end{pmatrix} - B\boldsymbol{v} \qquad T = \Sigma T_i$$

Gravity force

Total thrust



Quadrotor model 2/3

- **Rotational dynamics** due to pair-wise differences in rotor thrust:
  - Rolling torque

$$\tau_x = dT_4 - dT_2$$

$$\tau_x = db\left(\varpi_4^2 - \varpi_2^2\right)$$

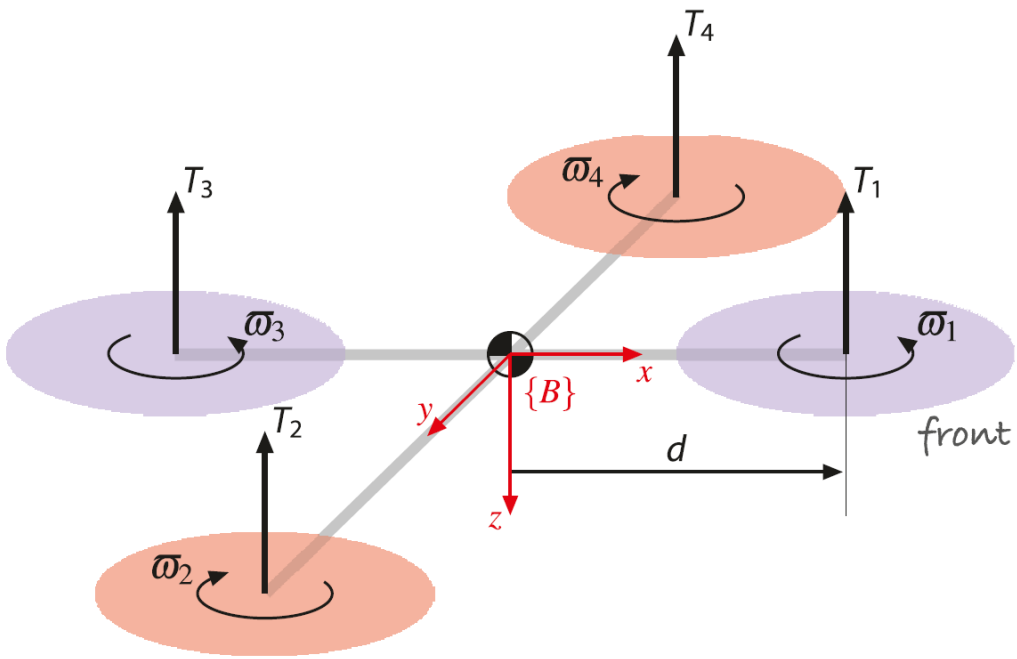
  - Pitching torque

$$\tau_y = db\left(\varpi_1^2 - \varpi_3^2\right)$$

  - Yaw torque:

$$\tau_z = Q_1 - Q_2 + Q_3 - Q_4$$

$$= k\left(\varpi_1^2 + \varpi_3^2 - \varpi_2^2 - \varpi_4^2\right)$$



Quadrotor model 3/3

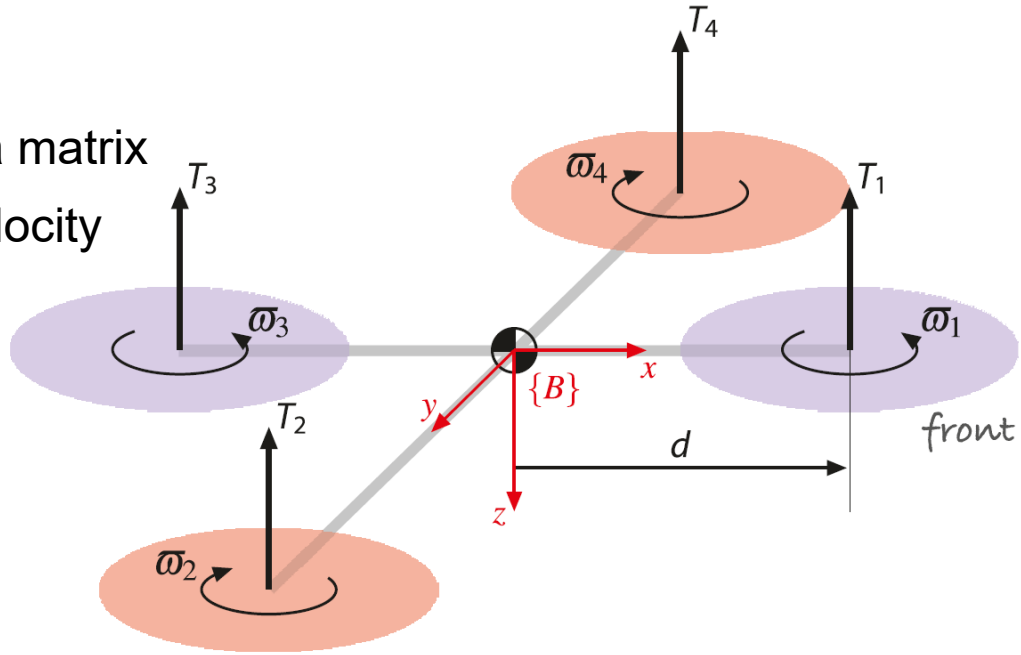
– Total torque: $\boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z)^T$

– Rotational acceleration (Euler's equation of motion):

$$J\dot{\omega} = -\omega \times J\omega + \tau$$

Intertia matrix

Angular velocity



Quadrotor control

- Under-actuated (4 actuators < 6 DoF), hence non-holonomic system
- Cannot move directly in x- (forward/backward) or y- (left/right) directions
- Must perform **maneuver** or sequence of motions to move in these directions (indirect translational motion)

Quadrotor Control – Target following

- ABB Open-house demo, Desember 2010
- Parrot AR Drone released at CES in January 2010 in LV.
- Student group discussion:S
  - Using regular PID-regulators, how would you design the control to achieve target following?

Quadrotor control: indirect translational motion (under-actuated)

- Cannot set pitch/roll angles directly
- Use a nested (cascade) control structure
- Outer loop: Position error gives required velocity:

$$ {}^{0}\boldsymbol{v}^{*} = K_p\left({}^{0}\boldsymbol{p}^{*} - {}^{0}\boldsymbol{p}^{\#}\right) $$

$$ \begin{pmatrix} {}^{B'}v_x \\ {}^{B'}v_y \end{pmatrix} = \begin{pmatrix} \cos\theta_y & -\sin\theta_y \\ \sin\theta_y & \cos\theta_y \end{pmatrix}^{T} \begin{pmatrix} v_x \\ v_y \end{pmatrix} $$

- And desired pitch angle:

$$ \theta_p^{*} \approx \frac{m}{T} K_f \left({}^{B'}v_x^{*} - {}^{B'}v_x^{\#} \right) $$

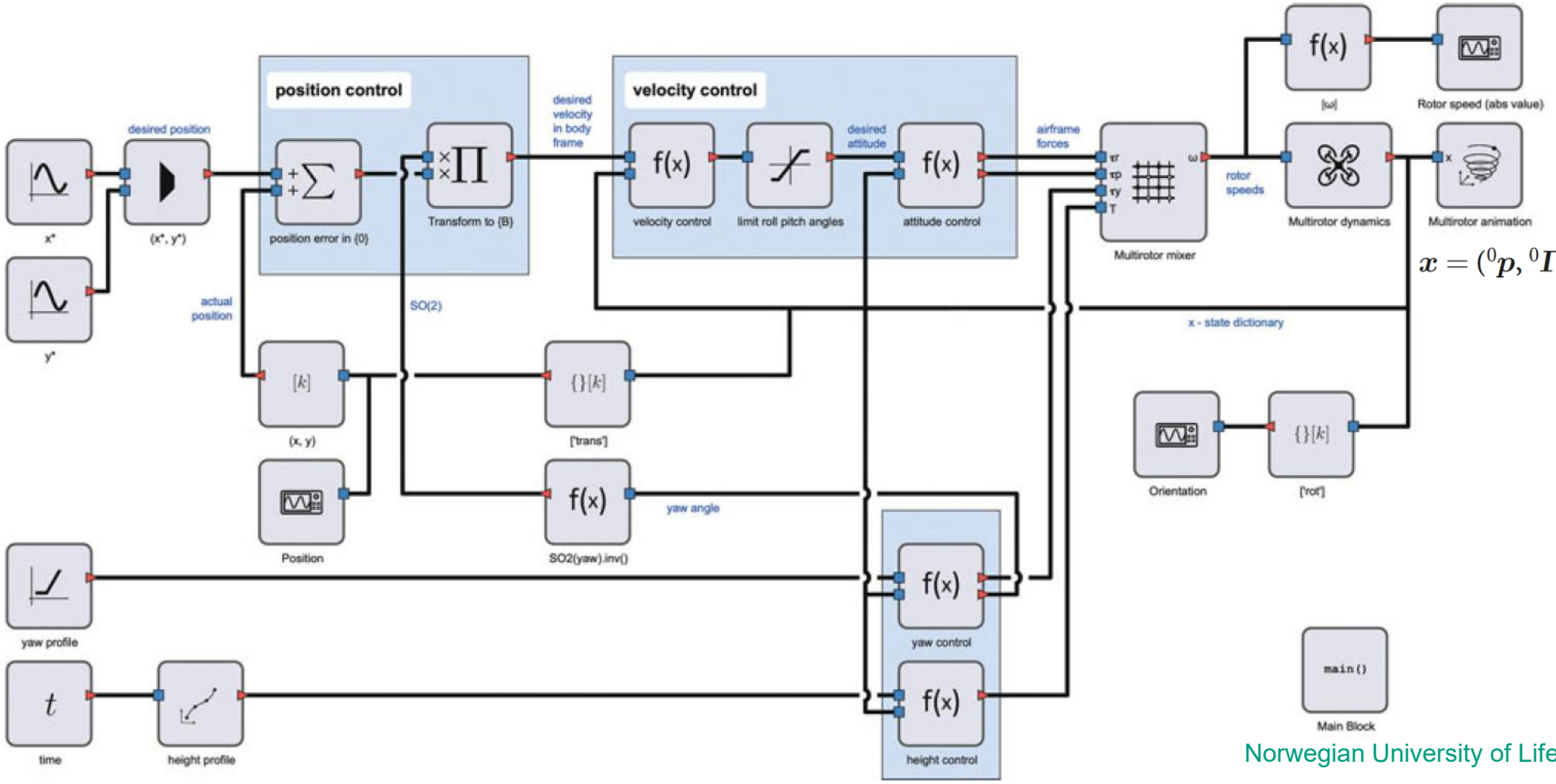
- Inner-loop: the rotational dynamics has a second order transfer function which requires a PD-controller:

$$ \tau_y^{*} = K_{\tau,p}\left(\theta_p^{*} - \theta_p^{\#}\right) + K_{\tau,d}\left(\dot{\theta}_p^{*} - \dot{\theta}_p^{\#}\right) $$

- Altitude controlled by PD-controller with T_0 = mg as feedforward term

$$ T = K_p\left(z^{*} - z^{\#}\right) + K_d\left(\dot{z}^{*} - \dot{z}^{\#}\right) + T_0 $$

Quadrotor control – models/quadroto



$$\boldsymbol{x} = ({}^{0}\boldsymbol{p}, {}^{0}\boldsymbol{\Gamma}, {}^{B}\dot{\boldsymbol{p}}, {}^{B}\dot{\boldsymbol{\Gamma}}) \in \mathbb{R}^{12}$$

Second project assignment: Walking Robot

- Consists of three parts as described in TEL200_WalkingRobot.pdf
    1. Motion planning
    2. Path planning
    3. Localization & Mapping

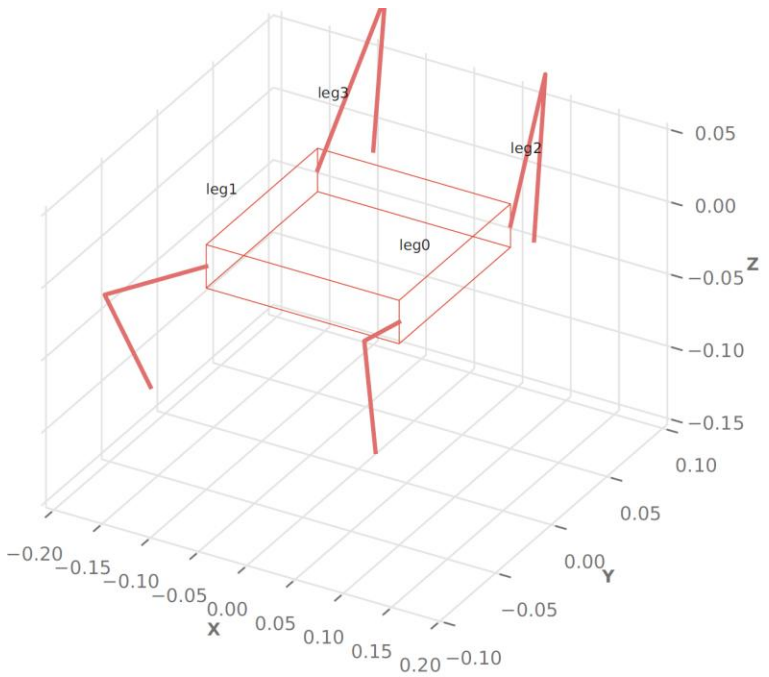


Figure 1: The walking robot, RVC Chapter 7.4.2

Second project assignment: Walking Robot

- Consists of three parts as described in TEL200_VT23_WalkingRobot.pdf
    1. Motion planning
    2. Path planning
    3. Localization & Mapping
- This project will count and constitute the remaining 40% of the overall course assessment.
- The project evaluation – which will be graded (A-E/F) - is based on the submitted files:
    1. written report (max 10 pages)
    2. video (maximum 3-minute-long)
    3. all Python files
- All files must be packed into in one single zip file and named "TEL200_Gr_X.zip" where X = your group number. Please make only one submission per group.
- **Submission deadline is set to 2026-04-30**

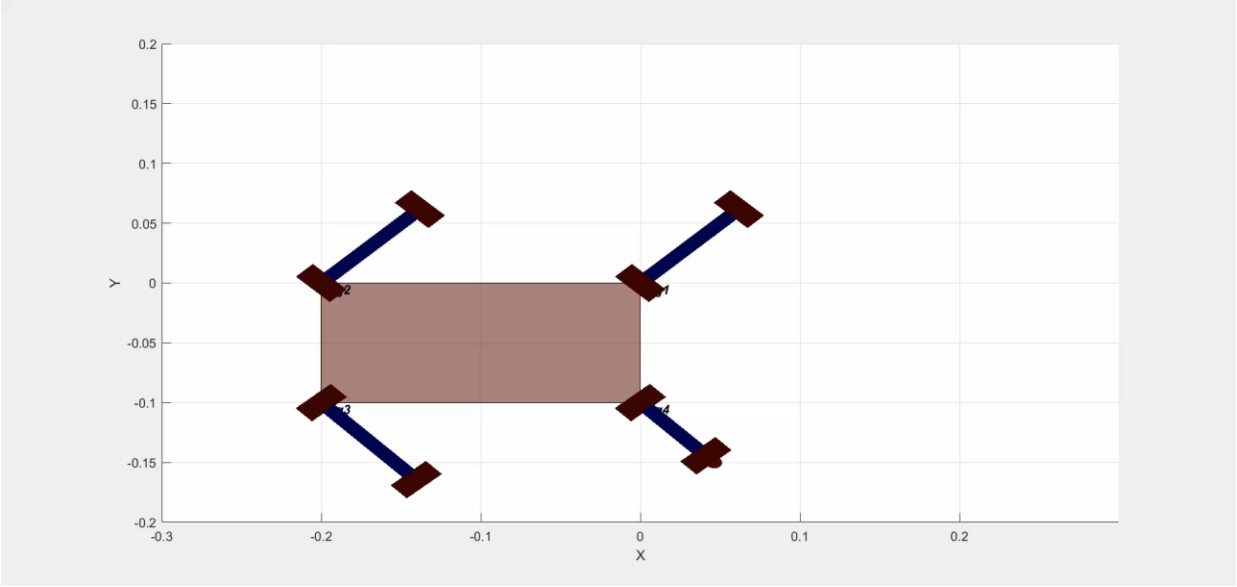


Figure 1: The walking robot, RVC Chapter 7.5.2

Thank you for your attention!